

ARNAU NECHES VILÀ

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EDUCATION

Universidad de Sevilla - Seville, Spain

Sep. 2022 - May 2027

Double Degree in Mathematics & Computer Science | GPA: 9.3/10 | 78+ ECTS with highest honours

Selected coursework: Real & Complex Analysis, Probability Theory, Differential Geometry, Abstract Algebra, Mathematical Programming, Algorithms & Data Structures, Artificial Intelligence, Operating Systems.

The University of Sydney - Sydney, Australia

Aug. 2025 - Jun. 2026

Exchange Student, Mathematics & Computer Science | WAM: 93/100

Selected coursework: Natural Language Processing, Deep Learning, Operating Systems Internals, Differential Geometry, Nonlinear ODEs with Applications, Complex Analysis.

RESEARCH EXPERIENCE

MINERVA AI Lab, Universidad de Sevilla - Undergraduate Researcher

Jul. 2025 - Present

- Developed and evaluated two new feature-selection methods designed to better capture interactions between features; led the research, implementation, and experiments across 6 synthetic and 4 real-world datasets.
- First author of "Lookahead and Dynamic Beam Search Variants of FOCI for Conditional-Dependence Feature Selection," accepted at CAEPIA 2026 and forthcoming in Lecture Notes in Artificial Intelligence (LNAI).
- Co-authored "Benchmarking Chronos-2 as a Foundation Model for Traffic Forecasting," also accepted at CAEPIA 2026 and forthcoming in LNAI.

PROFESSIONAL EXPERIENCE

Welyto - Lead Software Engineer

Apr. 2026 - Aug. 2026

Early-stage startup - conversational AI companion for older adults

- Led the software development of an AI-powered conversational device for older adults, spanning backend architecture, cloud infrastructure, AI integration, real-time device communication, and the family-facing application.
- Built the backend and device-cloud platform using Python, FastAPI, PostgreSQL, WebSockets, Google Cloud, Vertex AI / Gemini, and React, including persistent user context and family-facing functionality.
- Took the product from development to deployment with 50 paying beta families using physical devices, handling hundreds of real-world conversations, and contributed to the architecture of the next-generation product.

SELECTED PROJECT

Accessible Activity Data Platform - University of Sydney

13-week capstone

Project Manager & Developer | 7-person team | Institute of Musculoskeletal Health

- Led a 7-person team over a 13-week capstone project, acting as project manager, developer, and client liaison to build a platform helping people with disabilities discover suitable physical activities.
- Built an AI-powered data collection pipeline that discovered activity providers online and transformed unstructured web content into structured data; developed the database and web interface and integrated Gemini API and Google Maps API.

LEADERSHIP & ACTIVITIES

Competitive Programming Club, Universidad de Sevilla

2024 - Present

Events & Institutional Relations

- Active competitive programmer; qualified with my team for the national Ada Byron programming contest in both 2024 and 2025.
- Organize academic and industry events and manage external relations, building relationships with 10+ companies for sponsorships, talks, and collaborations.

Sputnik / Los 300

2025

- Selected for Los 300 after completing Sputnik, a selective programme bringing together young people from southern Spain around technology, entrepreneurship, innovation, leadership, and social impact.

TECHNICAL SKILLS

Programming: Python, C, Java, JavaScript, SQL, MATLAB

ML & Data: PyTorch, scikit-learn, Pandas, NumPy

Software: FastAPI, Flask, React, PostgreSQL, SQLAlchemy, WebSockets, REST APIs

Cloud & Tools: Google Cloud Platform, Docker, Git, Linux

LANGUAGES

Spanish: Native | **Catalan:** Native | **English:** C1 (Cambridge Advanced, Grade A)